

character cannot be developed in ease & quiet. only through experience of trial & suffering can the soul be strengthened
—Helen Keller

PARTIAL FRACTIONS:

To resolve a single rational fraction as a sum of two or more fractions is called partial fraction resolution.

Rational fraction: A fraction $\frac{P(x)}{Q(x)}$ (where P & Q are polynomial) is called a rational fraction if both $P(x)$ & $Q(x)$ are polynomials & $Q(x) \neq 0$.

e.g. $\frac{x^3 + 2x^2 - 3}{x+1}$ is a rational fraction.

$\frac{\sqrt{x}+1}{x^2+3}$ (not a polynomial) is not a rational fraction.

Proper fraction: if $\deg(P(x)) < \deg(Q(x)) \Rightarrow \frac{P(x)}{Q(x)}$ is a proper fraction.
e.g. $\frac{x+1}{x^2-3x+5}$

Improper fraction: if $\deg(P(x)) \geq \deg(Q(x)) \Rightarrow \frac{P(x)}{Q(x)}$ is improper fraction.
if two numbers fraction < 1
 $\hookrightarrow$ Proper fraction

e.g. $\frac{x+3}{x^2+2x+1}$, $\frac{x^3+x^2-1}{x^2+5}$
if two numbers fraction ≥ 1
 $\hookrightarrow$ Improper fraction.

Conditional equation: An equation which is true for some particular values of the variable involved is called conditional equation.

e.g. $x-2=0$ if $x=2$ 2 ke ilawa yeh kisi aur value ke liye true nah hai.

e.g. $x^2-9=0$ if $x=-3, 3$ (only true for 2 value).

Identity: An equation which is true for all values of the variable in the domain is called identity.

e.g. $\sin^2 x + \cos^2 x = 1 \quad \forall x \in \mathbb{R}$
 $(x+2)^2 = x^2 + 4x + 4 \quad \forall x \in \mathbb{R}$

Apni domain ki har value ke liye true ho.
aur hum identity kahein gein.

Note: When a rational fraction is separated is separated into sum of two or more fractions, then the result is a/an identity.

$$\frac{2}{x+1} + \frac{1}{x+3} = \frac{3x+7}{(x+1)(x+3)}$$

Partial fraction (resolve fractions)

L.C.M (combine fractions)

NOTE: Partial fraction is basically reverse process of taking L.C.M.

MCQ (1) $\frac{3x+7}{(x+1)(x+3)} = ?$

(a) $\frac{1}{x+1} + \frac{2}{x+3}$

(b) $\frac{2}{x+1} + \frac{3}{x+3}$

(c) $\frac{2}{x+1} + \frac{1}{x+3}$

(d) $\frac{2}{x+1} - \frac{1}{x+3}$

To solve these types of mcqs put any easy value of x & first put it in q & then in all the options which are given. if any of answer matches with question answer, that will be your correct option.

Put $x=0$

$\frac{3(0)+7}{3} = \frac{7}{3}$ that matches with op. (c)

NOTE: Results of partial fraction is always identity.

CASES:

case (i) $\frac{P(x)}{(Linear\ factors)} ; \frac{x+1}{(x+4)(x-3)}$

case (ii) $\frac{P(x)}{(Repeated\ linear\ factors)} ; \frac{x+1}{(x+1)^2(x-3)^3}$

case (iii) $\frac{P(x)}{(Irreducible\ Quadratic\ factors)} ; \frac{x+1}{(x^2+x+1)(x^2+1)}$
 $\rightarrow$ can't convert into linear.

case (iv) $\frac{P(x)}{(Repeated\ Irreducible\ Quadratic\ factors)} ; \frac{x+3}{(x^2+1)^2}$

$x^2-1 = (x-1)(x+1)$
 $\rightarrow$ reducible Quadratic factor equation. (reducible to linear).

$x^2+1 \neq (x-1)(x+1)$
 $\rightarrow$ irreducible Quadratic factor. (not reducible to linear)

→ mixture of case (i) & case (ii)

$$M(Q) \# (02) \quad \frac{x^2}{(x-2)(x-1)^2} = ?$$

don't put any value that make denominator = 0.

Put $x=0$

$$(a) \quad \frac{4}{(x-2)} + \frac{3}{(x-1)} + \frac{1}{(x-1)^2} \quad -2-3+1 \neq 6 \quad \frac{(0)^2}{(-2)(-1)^2} = 0$$

$$(b) \quad \frac{4}{(x-2)} - \frac{3}{(x-1)} - \frac{1}{(x-1)^2} \quad -2+3-1=0$$

$$(c) \quad \frac{1}{(x-2)} + \frac{2}{(x-1)} + \frac{3}{(x-1)^2} \quad \frac{-1-2+3}{-1-4+6} = \frac{1}{2} \neq 0$$

(d) None.

$$M(Q) \# (03) \quad \frac{x+1}{(x+3)(x+2)(x+4)} = \frac{A}{(x+2)} + \frac{B}{(x+3)} + \frac{C}{(x+4)}$$

Find value of A, B & C.

To find value of A put $x+2=0$, $x=-2$ in eq.

$$\frac{-2+1}{(-2+3)(-2+4)} \text{ ignore } (x+2) \quad = \frac{-1}{2} \quad , \quad A = -1/2$$

To find value of B put $x+3=0$, $x=-3$ in eq.

$$\frac{-3+1}{(-3+2)(-3+4)} \text{ ignore } (x+3) \quad = \frac{-2}{(-1)(1)} = 2 \quad , \quad B = 2$$

To find value of C put $x+4=0$, $x=-4$ in eq.

$$\frac{-4+1}{(-4+3)(-4+2)} \text{ ignore } (x+4) \quad = \frac{-3}{(-1)(-2)} = \frac{-3}{2} \quad , \quad C = -3/2$$

Now,

$$\frac{x+1}{(x+3)(x+2)(x+4)} = \frac{-1}{2(x+2)} + \frac{2}{(x+3)} - \frac{3}{2(x+4)}$$

Note:- Whenever difference is same of factors in denominator then

Sum of A, B & C will be equal to zero.

$$\frac{x+1}{(x+3)(x+2)(x+4)} = \frac{-1}{2(x+2)} + \frac{2}{(x+3)} - \frac{3}{2(x+4)}$$

$$A+B+C = \frac{-1}{2} + \frac{2}{2} - \frac{3}{2} = \frac{-1+4-3}{2} = \frac{-4+4}{2} = 0 \text{ hence proved! } (3)$$

M(Q# 04) $\frac{8x-11}{(x-1)(x-2)} = \frac{A}{(x-1)} + \frac{B}{(x-2)}$

what's value of A & B?

$A = 3$

$B = 5$

$\frac{8x-11}{(x-1)(x-2)} = \frac{3}{(x-1)} + \frac{5}{(x-2)}$

M(Q# 05) $\sum_{k=1}^{10} \frac{1}{k(k+1)} = ?$

if there exist consecutive products they break like $\frac{1}{4(5)} = \frac{1}{4} - \frac{1}{5}$

~~$\frac{1}{k(k+1)}$~~ $\frac{1}{k} - \frac{1}{k+1}$

$(\frac{1}{1} - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \dots + (\frac{1}{10} - \frac{1}{11})$

$= 1 - \frac{1}{11}$

$= \frac{10}{11}$

M(Q# 06) $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = (\frac{1}{1} - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots$
 $= 1$ only 1st term will be left

M(Q# 07) $\frac{1}{k(k+1)} = \frac{1}{1} (\frac{1}{k} - \frac{1}{k+1})$ as difference was one.

if $\frac{1}{k(k+1)(k+2)} = \frac{1}{2} (\frac{1}{k+1} - \frac{1}{k+2})$

→ now difference is (2) that's why.

if $\frac{1}{(k+2)(k+3)} = \frac{1}{3} (\frac{1}{k+2} - \frac{1}{k+3})$ difference is (3):

NOTE: In a polynomial, the variable can have only non-negative integer powers, while constants can be any numbers, including fractions, negatives or roots.
 • Fractional powers (like $x^{1/2}, x^{3/4}$) or negative powers like (x^{-1}, x^{-2}) of the variable are not allow.
 • constants, however, can be fractions, negatives or roots because they are just numbers not variables.

Hyperbolic Functions:

Where $e \approx 2.71$

e^x cannot be zero or negative

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \begin{cases} \text{Domain} = \mathbb{R} \\ \text{Range} = \mathbb{R} \end{cases}$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \begin{cases} \text{Domain} = \mathbb{R} \\ \text{Range} = [1, +\infty) \end{cases}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \text{Domain} = \mathbb{R}$$

$y = mx + c$ tangent to $x^2 + y^2 = a^2$

(4)

$$c^2 = a^2(1 + m^2)$$

$$c = \pm a\sqrt{1 + m^2}$$

$y = mx + c$ is tangent to $y^2 = 4ax$

$$c = \frac{a}{m}$$

Hyperbolic Functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

Domain = $\mathbb{R}$

Range = $\mathbb{R}$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Domain = $\mathbb{R}$

Range = $[1, +\infty)$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Domain = $\mathbb{R}$

Where $e \approx 2.71$

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